

EXERCISE – Run Phase

Your target is **LockBit ransomware group** – the most prolific ransomware operation ever documented, dismantled by Operation Cronos in February 2024, with extensive public documentation available.

Using only public clearnet sources – Chainalysis blog, Europol press releases, FBI advisories, Krebs on Security, Recorded Future public reports – reconstruct:

1. LockBit's full operational lifecycle (emergence through disruption)
2. Their forum presence and recruitment methodology
3. The specific OPSEC failures that enabled attribution of LockBitSupp (the leader)
4. Evidence of rebrand attempts post-Operation Cronos
5. Tradecraft fingerprints that would identify LockBit affiliates operating under new names

Write your full analyst note.

What Happened?

Confidence: high

On February 20, 2024, the [U.K.'s National Crime Agency (NCA)] (<https://www.nationalcrimeagency.gov.uk/news/nca-leads-international-investigation-targeting-worlds-most-harmful-ransomware-group>), alongside the [U.S. Department of Justice (DOJ)] (<https://www.justice.gov/opa/pr/us-and-uk-disrupt-lockbit-ransomware-variant>), announced the disruption of Lockbit, which has been one of the most prolific Ransomware-as-a-Service (RaaS) groups operating over the last few years. In this operation, the NCA, FBI, and international law enforcement partners worked together to seize servers and public-facing websites that were integral to Lockbit's operations, and have obtained decryptor keys for Lockbit victims to recover their data without paying a ransom.

Source: Chainanalysis

OFAC notes ten cryptocurrency addresses in sanctions against Lockbit affiliates Artur Sungatov and Ivan Kondratyev

In its SDN List entries for Lockbit affiliates Ivan Kondratyev and Artur Sungatov, OFAC included a total of ten cryptocurrency addresses controlled by the two individuals — nine for Kondratyev and one for Sungatov. Those addresses are below.

Ivan Kondratyev cryptocurrency addresses

- 1A7SKE2dQtezLktCY8peLsdAtkqxV9r1dC (Bitcoin)
- Bc1q8ew45w2agdfjfrnwp6adt2gqrc9n4mkev9ns29c (Bitcoin)
- bc1qagp0gy58v8hqvw4p2wsphcxg067rrppp45hexr (Bitcoin)
- bc1qn6segn8km4nfdp9vueu6msfjsaxaqgun9h60n9 (Bitcoin)
- bc1qx9upga7f09tsetqf78wa3qrmcjar58mkwz6ng6 (Bitcoin)
- 0xf3701f445b6bdafedbca97d1e477357839e4120d (Ethereum)
- 15cRqR3TXS1JehBGWERuxFE8NhWZzfoeeU (Bitcoin)
- bc1q5jqgm7nvrhaw2rh2vk0dk8e4gg5g373g0vz07r (Bitcoin)
- 32pTjxTNi7snk8sodrgfmdKao3DEn1nVJM (Bitcoin)

Artur Sungatov cryptocurrency address

- 18gaXypKj9M23S2zT9qZfL9iPbLFM372Q5 (Bitcoin)

A Dive Into Those Addresses

Source: Blockchain

Knowing all these addresses probably uses a money laundry method, i've decided to just stop at these addresses overview without looking outputs, and other wallets i've also ignored all the "invalid" ones to save time

#first 1A7SKE2dQtezLktCY8peLsdAtkqxV9r1dC



1A7SK-9r1dC

USD

Base58 (P2PKH)



Bitcoin Address

1A7SKE2dQtezLktCY8peLsdAtkqxV9r1dC

Bitcoin Balance

0.00000000 • \$0.00

Summary

This address has transacted 2 times on the Bitcoin blockchain. It has received a total of 0.81500000 BTC \$65,901.90 and has sent a total of 0.81500000 BTC \$65,901.90. The current value of this address is 0.00000000 BTC \$0.00.

Total Received ●

0.81500000 BTC
\$65,901.90

Transactions ●

2

Total Sent ●

0.81500000 BTC
\$65,901.90

Total Volume ●

1.63 BTC
\$131,803

Info:

This address has transacted 2 times on the Bitcoin blockchain. It has received a total of 0.81500000 BTC \$65,881.41 and has sent a total of 0.81500000 BTC \$65,881.41. The current value of this address is 0.00000000 BTC \$0.00.

#second 15cRqR3TXS1JehBGWERuxFE8NhWZzfoeeU



15cRq-foeeU

USD

Base58 (P2PKH)



Bitcoin Address

15cRqR3TXS1JehBGWERuxFE8NhWZzfoeeU

Bitcoin Balance

0.00000000 • \$0.00

Summary

This address has transacted 2 times on the Bitcoin blockchain. It has received a total of 0.03173807 BTC \$2,565.42 and has sent a total of 0.03173807 BTC \$2,565.42. The current value of this address is 0.00000000 BTC \$0.00.

Total Received ●

0.03173807 BTC
\$2,565.42

Transactions ●

2

Total Sent ●

0.03173807 BTC
\$2,565.42

Total Volume ●

0.06347614 BTC
\$5,130.85

Info:

This address has transacted 2 times on the Bitcoin blockchain. It has received a total of 0.03173807 BTC \$2,566.63 and has sent a total of 0.03173807 BTC \$2,566.63. The current value of this address is 0.00000000 BTC \$0.00.

#third 32pTjxTNi7snk8sodrgfmdKao3DEn1nVJM



32pTj-1nVJM

USD

Base58 (P2SH)



Bitcoin Address

32pTjxTNi7snk8sodrgfmdKao3DEn1nVJM

Bitcoin Balance

0.00000000 • \$0.00

Wallet

Chart

Summary

This address has transacted 53 times on the Bitcoin blockchain. It has received a total of 52.78923274 BTC \$4,267,012 and has sent a total of 52.78923274 BTC \$4,267,012 The current value of this address is 0.00000000 BTC \$0.00.

Total Received ●

52.78923274 BTC

\$4,267,012

Transactions ●

53

Total Sent ●

52.78923274 BTC

\$4,267,012

Total Volume ●

105.57846548 BTC

\$8,534,025

Info:

This address has transacted 53 times on the Bitcoin blockchain. It has received a total of 52.78923274 BTC \$4,269,017 and has sent a total of 52.78923274 BTC \$4,269,017 The current value of this address is 0.00000000 BTC \$0.00.

#last 18gaXypKj9M23S2zT9qZfL9iPbLFM372Q5



18gaX-372Q5

USD

Base58 (P2PKH)



Bitcoin Address

18gaXypKj9M23S2zT9qZfL9iPbLFM372Q5

Bitcoin Balance

0.00000000 • \$0.00

Wallet

Chart

Summary

This address has transacted 567 times on the Bitcoin blockchain. It has received a total of 91.48374766 BTC \$7,396,162 and has sent a total of 91.48374766 BTC \$7,396,162 The current value of this address is 0.00000000 BTC \$0.00.

Total Received ●

91.48374766 BTC

\$7,396,162

Transactions ●

567

Total Sent ●

91.48374766 BTC

\$7,396,162

Total Volume ●

182.96749532 BTC

\$14,792,325

Info:

This address has transacted 567 times on the Bitcoin blockchain. It has received a total of 91.48374766 BTC \$7,397,485 and has sent a total of 91.48374766 BTC \$7,397,485 The current value of this address is 0.00000000 BTC \$0.00.

Emergence

Confidence: high

LockBit emerged in late 2019 as the **ABCD** ransomware variant before evolving into one of the world's most prolific ransomware-as-a-service (RaaS) syndicates, responsible for nearly **2,300 attacks** and over **\$120 million** in extorted payments. The group operated by licensing its malware to affiliates and aggressively scaling through a professionalized, global affiliate network that targeted high-value sectors like healthcare and finance.

Recruitment methodology

Confidence: high

The LockBit ransomware group primarily operates through underground forums such as **XSS** and **RAMP** to recruit affiliates and manage operations. While the group was banned from the XSS forum in early 2024 following a dispute with a user, it attempted to restore its presence there in September 2025 by voting to lift the suspension of its support account, though the vote failed.

Key forum-related activities and status include:

- **Recruitment and RaaS Model:** LockBit utilizes these forums to advertise its Ransomware-as-a-Service (RaaS) model, recruit affiliates, and distribute updates for variants like LockBit 3.0 and 5.0.
- **Operational Disruption:** In February 2024, international law enforcement seized LockBit's websites, and the group faced internal conflicts, including a ban from the XSS forum that lasted until late 2025.
- **Recent Resurgence:** After announcing LockBit 5.0 on the RAMP forum in September 2025, the group resumed activity and launched a new data leak site in December 2025, claiming victims despite remaining banned from key forums like XSS.

Rebrand Attempts

Confidence: high

Following **Operation Cronos** in February 2024, the LockBit group faced significant disruption, leading to speculation and evidence of potential rebranding or resurrection attempts under new names or versions.

- **LockBit-NG-Dev:** Trend Micro identified an in-development version of the ransomware with a completely new codebase and a shift to a **.NET core**, indicating efforts to rebuild infrastructure after the original source code was seized.
- **LockBit 5.0:** In late 2024 and into 2025, a new onion portal branded as **LockBit 5.0** appeared, fueling speculation of a resurrection; however, analysts suggest this may represent an **opportunistic rebranding** by third-party actors or a **honeypot** rather than a confirmed return of the original core team.
- **SuperBlack:** In March 2025, researchers reported that a group named **Mora_001** used a variant of LockBit 3.0, removing LockBit branding and adding custom modules to create a new strain called **SuperBlack**, suggesting a fragmentation or mutation of the original codebase.
- **Reposting Victims:** Post-disruption activity was largely characterized by LockBit republishing **old victim data** to inflate its victim count and maintain the appearance of operational continuity, rather than launching new, distinct campaigns.
- **Affiliate Dispersal:** The severe reputational damage and loss of trust led many affiliates to disperse to other groups or reconsider the viability of the LockBit brand, with experts noting that **rebranding** is a common tactic for ransomware gangs after major takedowns.

LockBitSupp Opsec Failures

Confidence: high

LockBitSupp's operational security (OPSEC) was critically compromised by operational mistakes that allowed law enforcement and independent researchers to identify him as **Dmitry Yuryevich Khoroshev**. Despite frequently boasting about his anonymity and OPSEC skills, Khoroshev made significant errors, such as accessing the internet without a VPN in 2022 and maintaining active social media accounts linked to his real identity.

Key OPSEC failures included:

- **Unmasked IP Addresses:** Khoroshev occasionally connected to LockBit infrastructure without using a VPN, revealing his authentic IP address.
- **Social Media Exposure:** Researchers identified VKontakte (VK) accounts linked to Khoroshev, which included his real photograph, undermining his anonymous persona.
- **Law Enforcement Access:** During **Operation Cronos**, authorities gained access to LockBit's backend and administrative panels, collecting IP addresses, negotiation logs, and cryptocurrency wallet data that further exposed the group's operations.

In response to these vulnerabilities and the exposure of his identity, LockBitSupp initially attempted to bolster OPSEC by offering a **\$1 million bounty** (later raised to **\$10 million**) for information leading to the identification of his own identity, ostensibly to

deter detection. The group also decentralized its affiliate panel across multiple servers to mitigate future risks from law enforcement seizures.

Tradecraft Fingerprint

Source: wikipedia

LockBit operators frequently gain initial access by exploiting vulnerable [Remote Desktop Protocol](#) (RDP) servers or compromised credentials purchased from affiliates. Initial access vectors also include [phishing](#) emails with malicious attachments or links, [brute-forcing](#) weak RDP or [VPN](#) passwords, and exploiting vulnerabilities such as CVE-2018-13379 in [Fortinet](#) VPNs.¹[1\(https://en.wikipedia.org/wiki/LockBit#cite_note-:0-1\)](https://en.wikipedia.org/wiki/LockBit#cite_note-:0-1)

Once installed, LockBit ransomware is often executed in [Microsoft Windows](#) via [command-line arguments](#), [scheduled tasks](#), or [PowerShell scripts](#) such as PowerShell Empire. LockBit uses tools such as [Mimikatz](#), GMER, Process Hacker, and registry edits to gather [credentials](#), disable security products, and evade defenses. It enumerates network connections to identify high-value targets such as [domain controllers](#) using scanners such as Advanced Port Scanner.¹[1\(https://en.wikipedia.org/wiki/LockBit#cite_note-:0-1\)](https://en.wikipedia.org/wiki/LockBit#cite_note-:0-1)

For [lateral movement](#), LockBit spreads through [SMB file-sharing](#) connections inside networks, using credentials gathered earlier. Other lateral movement techniques include distributing itself via compromised [Group Policy](#) objects, or using tools such as PsExec or [Cobalt Strike](#).¹[1\(https://en.wikipedia.org/wiki/LockBit#cite_note-:0-1\)](https://en.wikipedia.org/wiki/LockBit#cite_note-:0-1)

LockBit's ransomware payload encrypts files and network shares using [AES](#) and [RSA encryption](#). It encrypts only the first few [kilobytes](#) of each file for faster processing, and adds a ".lockbit" extension. LockBit then replaces the desktop wallpaper with a ransom note; it can also print ransom notes to attached printers. The goal is to extort payment of a ransom to reverse system disruption and restore file access.